Climate change is having an impact on Australian agriculture — from the effects of shifting weather patterns to the increasing market expectations for low emissions production. These challenges affect farmers, and the rural communities they live and work in.

There are also opportunities. Taking steps to manage your emissions can boost productivity over time, open new income streams and make your farm more resilient, all while helping the environment.

The Knowledge Hub is here to support you. Whether you're looking to understand the changes, take action, or find help, we're building a space to guide you through.

Get clear, practical information to help you feel confident and informed.

Watch

PROFESSOR RICHARD ECKARD: For farmers and land managers to meet the goal of reduced emissions starting in 2030 through to 2050, they need to know what to do next, what steps to do next, and they need to know where the policy environment is coming from, who's asking them to be low emissions, what the targets are, and then what the options are for them to start responding.

Hi. I'm Richard Eckard, professor in the Faculty of Science at the University of Melbourne. I lead the Primary Industries Climate Challenges Centre, which researches the impact of climate change on agriculture and agriculture on climate.

What we're seeing is all the multinational supply chain companies that deal with agricultural produce have set targets, targets for reduced greenhouse gas emissions. And they average somewhere around 30 percent less emissions by 2030 and net zero by 2050. What we also know is about 70 percent of Australian agricultural produce is exported down these multinational supply chain targets. And so how does Australia perform on the global stage when those companies start buying globally to meet their target?

So it's really imperative that farmers and land managers get on board to know how do they gear their system to deliver the low emissions product that the supply chain will want to buy by 2030. What we're trying to do is just bring up the knowledge that carbon farming is a part of their future.

There is this trajectory towards lower emissions. So making them aware of the policy environment, of the supply chain constraints, of how they need a partnership with their supply chain, to achieve this. And then some awareness of what is their number, how do they get their number, and how do they move down the track towards improving that number. And what are the technologies they can bring to bear to reduce their number, their greenhouse gas footprint?

So these will be things to start with are just best practice. Best practice that we've known for the last 40 years. Things like nitrogen use efficiency, better crop yields, better soil testing, better growth rates in livestock, feeding animals better, bringing legumes into agriculture. These are all things we've known for a long time that improve efficiency, but also reduce the greenhouse gas footprint.

Australia is already 22 percent more rainfall variable than any other country in the world, and the historic management of the land took that into account. Now we're becoming aware of this in how we do carbon farming, that we have to actually change from strictly European farming systems to systems that are more attuned to this high variability we're encountering. And so there's a lot to be learned from the Indigenous land management practices that we need to then incorporate into traditional farming, non Indigenous farming, so that it actually is a bit more in tune with the high variability we have in Australia.

Now the world needs to go net zero by 2050. What we haven't really reconciled is where does the big emission reduction take place? Obviously, it has to happen in the fossil fuel sector.

But we need to move towards, well, what can agriculture contribute to that inevitable net zero? And what can they contribute towards the 2030 goal? Now not every agricultural sector has the identical opportunity. We've got some intensive horticulture for example that have very low emissions and almost nothing to do to get to net zero apart from renewable energy. But you've got an extensive livestock sector where a lot of northern cattle stations, we don't even know how many cattle are there. So the challenges are vastly different, and this is what the program is trying to address is who has what options to move forward and what are those options.

About the Knowledge Hub

The Australian Government is working with industry to encourage climate-smart practices to support a sustainable and productive agriculture sector. Access to trusted sources of information is a priority of the Agriculture and Land Sector Plan, which was developed in consultation with industry.

Emissions management is an emerging and complex area. Having access to trusted sources of information can help farmers make informed decision about emissions management.

The Knowledge Hub is one of several activities under the Carbon Farming Outreach Program. It aims to help farmers learn about emissions management, and how to use that knowledge on their farms.

Share your feedback

We are still developing the Knowledge Hub. We will add more content as we expand on the current topics and add new ones.

In the meantime, we’re interested in your feedback to help build an effective Hub. Please use the ‘Was this page helpful’ feature to share your feedback.

Source: https://www.agriculture.gov.au/agriculture-land/farm-food-drought/climatechange/carbon-farming-outreach-program/hub